

UMANG J GALA - TECHNICAL ARCHITECT

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SKILLS

Programming

- Java • Typescript • Javascript
- Python • C/C++ • PHP
- HTML 5 • CSS
- SQL • Shell / Bash

Framework & Libraries

- React • Angular
- Kafka • Zookeeper
- Spark - Stream / SQL • Hadoop
- Avro • Protobuf
- Spring • PJ 2 • JUnit

Platform

- Azure - AKS, Storage, EventHub, Networking, etc
- AWS - EC2, Lambda, S3, DDB, SQS, SNS, Cloudwatch, etc
- Kubernetes & Docker

Software Tools

- Node.js • Gradle • Maven
- AWS CLI • Jenkins
- GIT • GitHub
- VS Code • IntelliJ Idea

OS

- Mac OS • Windows • Linux

Soft Skills

Leadership
Project Management
Team Management
Problem-Solving

CERTIFICATION

Cloud

AWS Cloud Practitioner
Microsoft Azure Fundamentals

EDUCATION

Rochester Institute of Technology

M.S. in CS

CGPA - 3.29

May 2015 | Rochester, NY

K J Somaiya College of Engineering

B.E. in IT

Aggregate - 66%

May 2013 | Mumbai, India

EXPERIENCE

Microsoft | Senior Software Engineer | Technical Architect

Nov 2024 - Present | Noida, India

- **Core Responsibilities & Ownership** : Architected and delivered high-scale backend services for Microsoft Purview Data Map, ensuring resilient ingestion pipelines, optimized metadata storage, and consistent performance at enterprise scale.
- **Technical Impact** : Drove deep technical improvements in distributed backend systems, eliminating scalability bottlenecks and strengthening reliability for high-throughput enterprise metadata services
- **Customer & Business Value** : Delivered customer-centric backend features for Microsoft Purview Data Map, translating enterprise governance needs into reliable, scalable capabilities that improved metadata discoverability and trust.
- **Cross-Team Collaboration** : Drove cross-team collaboration across scanning, catalog, and MIP/RBAC teams, working closely with PM, design, and partner engineers to translate customer requirements into scalable, production-ready solutions across the Purview ecosystem.
- **Security, Quality & Compliance** : Owned SFI-driven quality and security improvements for the service, strengthening CI/CD, test coverage, and SDL compliance to meet Microsoft engineering standards.
- **Leadership & Mentorship** : Provided technical mentorship to junior engineers, driving best practices in system design and distributed systems, and fostering a culture of engineering excellence through documentation, knowledge sharing, and post-incident learning. partnered with JP Morgan

Amazon | Technical Architect & Designer

May 2021 - Feb 2023 | Hyderabad, India

- Architected an end-to-end data labeling platform used by 4 engineering teams, enabling ML scientists to train AI models with image, video, and text data. (Java, Python, React, AWS)
- Designed a secure microservice architecture to protect Alexa customer data; this privacy-enforcing pattern was mandated and adopted as the standard across all org-wide engineering teams. (Java, AWS)
- Developed a modular UI framework with a plugin system adopted by 3 teams, streamlining development and component ownership in a large-scale web application. (React, HTML, CSS)
- Partnered with the AWS organization to co-develop and influence the roadmap for core machine learning platform features, impacting both internal and external services.
- Provided technical mentorship to new engineers, accelerating their integration and driving best practices in system design and code quality.
- Championed Agile methodologies and upheld strict SDLC standards, ensuring all projects adhered to rigorous security, privacy, and IT controls.

JP Morgan Chase & Co. | Senior Software Engineer

Feb 2019 - May 2021 | Newark, DE

- Designed and developed a distributed data lake application purely driven by metadata for easy usability and scalability (Kubernetes, Spark, Java, Kafka, Avro, Protobuf)
- Designed and developed a Real-Time Model Transformation application that converts any input data model to a specified output model with data lineage (Spark Stream/SQL, Java, Avro)
- Development and test distributed application changes using disciplined software development processes (Java, Kafka, Zookeeper, Avro)
- Day-to-day maintenance of the application system in operations, including tasks related to identifying and troubleshooting application issues and issues resolution or escalation and catering to ad-hoc work requests
- Adherence to Development Life Cycle, IT controls and change control policies for all developments
- Mastered software engineer using Agile methodology by participating in product backlog grooming, sprint planning, review and retrospective

COURSEWORK

Graduate

Parallel Computing
Data Security & Privacy
Distributed Systems
Computer Networks
Cryptography
Neural Networks
Machine Learning
Big Data
Algorithms

Undergraduate

Information & Network Security
Software Engineering
Object Oriented Programming
Data Structures
Database Technologies
Microcontrollers and processors

PERSONAL

INTERESTS

Reading
Writing
Travelling
Volunteering
Photography

EXPERIENCE

JP Morgan Chase & Co. | Software Engineer

Feb 2015 - Jan 2019 | Newark, DE

- Developed full-stack features using Java (Spring, REST) and Angular, including a metadata-driven data browser and a scalable, configuration-based data landing framework.
- Improved system security and insights by building an automated vulnerability scanner (Bash) and developing various data reports using ETL tools (Pentaho, Oracle).
- Demonstrated leadership by mentoring peers to promote code reuse and volunteering as Tech Lead to successfully deliver a web application for a non-profit partner.

JP Morgan Chase & Co. | Software Engineer - Intern

Jun 2014 - Aug 2014 | Newark, DE

- Developed a full-stack back-office module (Java, Spring, ExtJs) and optimized a data processing program to handle hundreds of thousands of entries efficiently.
- Built an automated testing infrastructure using Selenium, including automatic test report generation and maintenance of the virtual testing environment (Unix).

Rochester Institute of Technology | GA - System Admin, CS Department

Aug 2014 - Apr 2015 | Rochester, NY

- Managed CS department infrastructure, including server/lab maintenance, user account administration, and Hadoop cluster setup; provided real-time technical support and developed automation scripts (Unix, Bash) to monitor resources.

PROJECTS

Financial Analysis Engine Sep 2025 - Present

Designed and developing an intelligent MCP server to answer complex financial queries by orchestrating multiple data models, seamlessly integrating historical and real-time data to perform unified analytics.

Scalable Generic Data Lake Platform Oct 2024 - Present

Architected a highly scalable, metadata-driven data lake platform capable of serving diverse domains such as finance, e-learning, and content management, establishing a generic framework for rapid data integration and analytics.

Indoor Localization Aug 2014 - May 2015

A mobile application that locate an object or a people inside a building using Received Signal Strength (RSS) from already deployed WiFi access point, as GPS is not suitable for indoor condition. It also help getting direction in huge indoor environments like schools, malls, airports, etc. Technologies used Android, Java & C++

Hybrid Parallel Logistic Regression Sep 2014 - Nov 2014

Used the Parallel Java 2 library to implement parallel logistic regression for machine learning. Ran experiments on a 10-node cluster parallel computer using very large and very sparse data-sets (3 million records against 2 million features). Technologies used Java, PJ2 Java library

Stock Recommendation Model Feb 2014 - Apr 2014

A system that recommends the stocks to users based on the current stocks in their portfolio. The system uses k means clustering algorithm on historical NASDAQ data to cluster the stocks. Technologies used MySQL database, SQL, Weka & PHP

Real-Time Editing System Sep 2012 - Apr 2013

A groupware system that allows group of users to view and edit the one document at real time. A user can create or upload a document and invite other users to contribute to that document. It provides access privileges settings to the owner of the document. Technologies used PHP, HTML, CSS, JavaScript, JQuery, Node.js & SQL

Information Sharing Aug 2014 - May 2015

A web application where users can create groups and share information or documents with individual or groups. It has a real time chat room to conduct online meetings. Technologies used JSP, Servlet, Javascript, HTML, CSS and SQL

Trivial File Transfer Protocol Aug 2014 - Nov 2014

Implemented TFTP used to transfer files that runs on top of UDP. This TFTP protocol is based on RFC-1350. Technology used Java